

Mathematics Study Notes (MA401)

Academic Insight - Public Sample Revision Summary

Module 1: Eigenvalues, Eigenvectors & Matrix Diagonalization

- Eigenvalue equation: $A v = \lambda v \iff \det(A - \lambda I) = 0$
- Trace identity: sum of eigenvalues = $\text{tr}(A)$; Determinant = product of eigenvalues
- Worked Example: For $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$, $\lambda^2 - 7\lambda + 10 = 0 \Rightarrow \lambda = 5, 2$

Module 2: Conditional Probability & Bayes Theorem

- Bayes Formula: $P(B_i | A) = \frac{P(A | B_i) P(B_i)}{\sum_j P(A | B_j) P(B_j)}$
- Variance Identity: $\text{Var}(X) = E[X^2] - (E[X])^2$
- Note: Sample educational revision sheet provided by Academic Insight.